

~~CONFIDENTIAL//SI~~

UNITED STATES GOVERNMENT

memorandum

DATE: 26 March 1992

REPLY TO
ATTN OF:

[] Z441

(b)(3)-P.L. 86-36

SUBJECT:

KRYPTOS sculpture

TO:

[] Z44

~~(FOUO)~~ In response to a direct challenge from the Central Intelligence Agency, NSA analysts have successfully diagnosed and read a major portion of the KRYPTOS sculpture, located in the courtyard of CIA in Langley, Virginia. As of 1 Dec 92, the cipher has been divided into four sections, with complete analysis and decryption completed on the first 3 parts.

1. ~~(FOUO)~~ Cipher parts 1 and 2 were encrypted using a polyalphabetic substitution system. Part 1 employed 10 alphabets, while part 2 used 8 alphabets. The third section was encrypted using a route transposition on a width of 86.

2. ~~(FOUO)~~ The decrypted text of part 1: "Between subtle shading and the absense of light lies the nuance of iqlusion.[sic]"

3. ~~(FOUO)~~ The decrypted text of part 2: "It was totally invisible. How's that possible? They used the earth's magnetic field. The information was gathered and transmitted underground to an unknown location. Does Langley know about this? They should. It's buried out there somewhere. Who knows the exact location? Only W.W. This was his last message. Thirty-eight degrees, fifty-seven minutes, six point five seconds north. Seventy-seven degrees, eight minutes, forty-four seconds west. I.D. by rows."

4. ~~(FOUO)~~ The decrypted text of part 3: "Slowly, desperatly[sic], slowly, the remains of passage debris that encumbered the lower part of the doorway was removed. With trembling hands I made a tiny breach in the upper left hand corner, and then, widening the hole a little, I inserted the candle and peered in. The hot air escaping from the chamber caused the flame to flicker, but presently, details of the room within emerged from the mist. Can you see anything? Q[sic]"

~~(FOUO)~~ Although ideas abound, a successful break into part 4 has not been made, and analysts continue to work for a solution.

Approved for Release by NSA on 05-21-2013,
EOIA Case # 61191

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INTRODUCTION

~~(U)~~ (S) The following paper will take a technical look at the solution to a major portion of the KRYPTOS sculpture located in the courtyard of the Central Intelligence Agency in Langley, Virginia. Before starting on the technical details, let's take a quick look at the history of the sculpture, as well as a few comments from the sculptor.

~~(U)~~ (FOUO) In June 1988, a Fine Arts Commission project was announced by the CIA to acquire art work for the new CIA Headquarters building. When the selection process had been completed, the Director of Central Intelligence approved the proposal submitted by James Sanborn, a Washington area artist, to create a two-part sculpture at the west entrance to the new Headquarters building, and in the courtyard of the complex. In the fall of 1990 the work was unveiled at a dedication ceremony at the CIA.

~~(U)~~ (FOUO) According to Mr. Sanborn, "the stonework at the entrance and in the courtyard served two functions. First, it creates a natural framework for the project as a whole and is part of a landscaping scheme designed to recall the natural stone outcropping that existed on the site before the Agency, and that will endure as do mountains. Second, the tilted strata tell a story like pages of a document. Inserted between these stone "pages" is a flat copper sheet through which letters and symbols have been cut. This code, which includes certain ancient ciphers, begins as International Morse and increases in complexity as you move through the piece at the entrance and into the courtyard. Its placement in a geologic context reinforces the text's "hiddenness" as if it were a fossil or an image frozen in time."

~~(U)~~ (FOUO) This paper's purpose is to concentrate solely on the copper sheets located in the courtyard through which letters and question marks were cut out. It will look at the diagnosis, exploitation and eventual solution of the majority of the cipher contained in the sculpture.

THE KRYPTOS SCULPTURE

~~U (FOUO)~~ One half of the sculpture contains the following Vigenere square, which uses mixed sequences based on the keyword **KRYPTOS**.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	
A	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	
B	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	
C	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	
D	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	
E	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	
F	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	
G	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	
H	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	
I	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	
J	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	
K	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	
L	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	
M	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	
N	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	<u>L</u>
O	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	
P	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	
Q	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	
R	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	
S	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	
T	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	
U	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	
V	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	
W	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	
X	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	
Y	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	
Z	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	

~~(FOUO)~~ The extra letter "L" at the end of the 15th line is as it appears in the sculpture. This Vigenere square will turn out to play a key role in reading 2 of the 3 cipher sections exploited by NSA analysts.

~~SECRET~~

~~U (FOUO)~~ Following is the other half of the main sculpture. Line numbers and underlining have been added for reference purposes only, and are not a part of the sculpture.

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1  E M U F P H Z L R F A X Y U S D J K Z L D K R N S H G N F I V J
2  Y Q T Q U X Q B Q V Y U V L L T R E V J Y Q T M K Y R D M F D
3  V F P J U D E E H Z W E T Z Y V G W H K K Q E T G F Q J N C E
4  G G W H K K ? D Q M C P F Q Z D Q M M I A G P F X H Q R L G
5  T I M V M Z J A N Q L V K Q E D A G D V F R P J U N G E U N A
6  Q Z G Z L E C G Y U X U E E N J T B J L B Q C R T B J D F H J R R
7  Y I Z E T K Z E M V D U F K S J H K F W H K U W Q L S Z F T I
8  H H D D D U V H ? D W K B F U F P W N T D F I Y C U Q Z E R E
9  E V L D K F E Z M O Q Q J L T T U G S Y Q P F E U N L A V I D X
10 F L G G T E Z ? F K Z B S F D Q V G O G I P U F X H H D R K F
11 F H Q N T G P U A E C N U V P D J M Q C L Q U M U N E D F Q
12 E L Z Z V R R G K F F V O E E X B D M V P N F Q X E Z L G R E
13 D N Q F M P N Z G L F L P M R J Q Y A L M G N U V P D X V K P
14 D Q U M E B E D M H D A F M J G Z N U P L G E W J L L A E T G
15 E N D Y A H R O H N L S R H E O C P T E O I B I D Y S H N A I A
16 C H T N R E Y U L D S L L S L L N O H S N O S M R W X M N E
17 T P R N G A T I H N R A R P E S L N N E L E B L P I I A C A E
18 W M T W N D I T E E N R A H C T E N E U D R E T N H A E O E
19 T F O L S E D T I W E N H A E I O Y T E Y Q H E E N C T A Y C R
20 E I F T B R S P A M H H E W E N A T A M A T E G Y E E R L B
21 T E E F O A S F I O T U E T U A E O T O A R M A E E R T N R T I
22 B S E D D N I A A H T T M S T E W P I E R O A G R I E W F E B
23 A E C T D D H I L C E I H S I T E G O E A O S D D R Y D L O R I T
24 R K L M L E H A G T D H A R D P N E O H M G F M F E U H E
25 E C D M R I P F E I M E H N L S S T T R T V D O H W ? O B K R
26 U O X O G H U L B S O L I F B B W F L R V Q Q P R N G K S S O
27 T W T Q S J Q S S E K Z Z W A T J K L U D I A W I N F B N Y P
28 V T T M Z F P K W G D K Z X T J C D I G K U H U A U E K C A R

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~~U (FOUO)~~ An initial look revealed a number of oddities. First was the inclusion of question marks. It was not known if these were being used to signify a transition from one cipher system to another, or if they acted as punctuation for the plaintext. Immediately after the halfway point was the word **END** which may be a coincidence or it might refer to the end of some cryptosystem. This was a distinct possibility because the **Y**, **A**, and **R** that followed **END** were actually raised slightly when compared to the surrounding letters, perhaps signaling the beginning of a different cryptosystem. All of these peculiarities would eventually be explained through the reading of the majority of the cipher.

~~CONFIDENTIAL~~**DIAGNOSIS**

~~(S)~~ The initial diagnosis of this cipher revealed the probable use of at least 3 separate cryptographies. The main reason for this assumption was that beginning with line 15, and proceeding through to line 25, numerous analysts noticed that a frequency count of the letters observed would roughly match that of the English language. If lines 15-25 used some particular cryptosystem, then it was likely that another was used for lines 1-14 and yet another one used for lines 26-28. That would yield a probable minimum of 3 distinct cryptosystems in use.

~~(S)~~ A statistical analysis of the first section (lines 1-14) showed a particular roughness on a width of 8. The most common explanation for width roughness is that of a polyalphabetic substitution system. In such a system, a message is encrypted using multiple simple substitutions, employing each substitution in a predetermined order. In this particular case, the width of 8 is a probable indication of 8 cipher alphabets being used.

~~(S)~~ The second section was already partly diagnosed, based solely on analysts "eyeballing" the cipher. Statistical programs confirmed that this section had the characteristics of English plain text, though obviously mixed up in some manner. The most likely explanation for this is a transposition system, perhaps a keyed columnar transposition. In such a system, the plain text is inscribed horizontally into a matrix, normally a rectangle, and then the letters are extracted vertically, according to a predetermined sequence.

~~(S)~~ [REDACTED] the data also revealed a bias in the third section at an interval of 7. There is no specific cryptography that would give such a result, but a number of them could yield such results under certain circumstances, some of which will be discussed in detail in a later section. After solution of other parts, we discovered that this section was actually the fourth part of the problem. Read on for how this was discovered.

(b)(1)
(b)(3)-50 USC 403
(b)(3)-P.L. 86-36

~~CONFIDENTIAL~~

THE FIRST BREAKTHROUGH

~~54~~ (S) Under the hypothesis that the first section employed a polyalphabetic substitution with 8 alphabets, a frequency count was done for each alphabet in the cipher. It was assumed that the alphabets were used in a sequential order, i.e., alphabet #1 was used to encipher characters number 1, 9, 17, 25, etc.; Alphabet #2 was used for characters 2, 10, 18, 26, etc.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
-	-	4	2	2	2	2	2	2	1	1	1	-	8	2	3	2	8	2	-	1	1	3	-	1	4
2	1	-	3	5	4	3	4	-	5	1	4	2	-	-	2	3	3	1	1	3	1	3	-	1	2
3	-	-	3	1	-	7	3	1	3	2	1	1	2	-	-	10	2	1	3	1	3	1	1	4	1
1	3	-	3	1	2	-	-	1	2	7	4	-	4	-	5	2	-	-	4	7	2	-	2	1	3
3	1	1	11	3	7	2	-	2	2	-	2	3	1	-	3	3	-	1	-	3	1	-	1	2	2
1	-	-	4	7	9	1	1	1	1	1	6	2	-	-	-	4	1	-	1	4	-	1	3	1	5
-	2	-	1	6	4	2	3	1	-	-	3	6	1	-	1	1	1	1	5	2	9	-	2	-	3
-	1	2	1	6	3	7	3	-	2	6	1	5	1	1	2	4	-	-	2	3	3	-	-	1	-

~~54~~ (S) Using the above frequency counts, it was hoped that one could place the alphabet sequence right on top of each row, though at some offset, and get something that would at least roughly match what you might expect to see. This did not lead to a hoped for solution. Since the accompanying Vigenere Square was based on the keyword **KRYPTOS**, the frequency counts were next sorted based on the same keyword mixed sequence.

K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
1	8	1	3	-	2	2	-	-	4	2	2	2	2	2	2	1	1	-	8	2	1	1	3	-	4
1	3	1	2	1	-	1	2	1	-	3	5	4	3	4	-	5	4	2	-	3	3	1	3	-	2
2	2	4	-	3	-	1	3	-	-	3	1	-	7	3	1	3	1	1	2	10	1	3	1	1	1
7	-	1	5	4	-	-	1	3	-	3	1	2	-	-	1	2	4	-	4	2	7	2	-	2	3
-	-	2	3	-	-	1	3	1	1	11	3	7	2	-	2	2	2	3	1	3	3	1	-	1	2
1	1	1	-	1	-	1	1	-	-	4	7	9	1	1	1	1	6	2	-	4	4	-	1	3	5
-	1	-	1	5	-	1	-	2	-	1	6	4	2	3	1	-	3	6	1	1	2	9	-	2	3
6	-	1	2	2	1	-	-	1	2	1	6	3	7	3	-	2	1	5	1	4	3	3	-	-	-

~~54~~ (S) Using the frequency count this way, it appeared possible to place a keyword mixed alphabet on at least some of these rows. In the third row for example, the following appeared to be a good alignment.

2	2	4	-	3	-	1	3	-	-	3	1	-	7	3	1	3	1	1	2	10	1	3	1	1	1
L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J

~~SECRET~~

U (S) In the sixth row the following looked like a good alignment as well.

1 1 1 - 1 - 1 1 - - 4 7 9 1 1 1 1 6 2 - 4 4 - 1 3 5
Q U V W X Z K R Y P T O S A B C D E F G H I J L M N

L (S) Using some of these assumptions and cribbing in words where needed, a solution was effected. Following are the plain alphabet, the 8 cipher alphabets, and a decryption of the text that was readable. Note two items: 1) there is a repeating key of **ABSCISSA** under the index letter of **K** in the plain alphabet, and; 2) the readable text actually begins with the cipher letter **V** at the beginning of the third line of the sculpture.

P: KRYPTOSABCDEFGHIJLMNQUVWXZ
C1: ABCDEFGHIJLMNQUVWXZKRYPTOS
C2: BCDEFGHIJLMNQUVWXZKRYPTOSA
C3: SABCDEFHGHIJLMNQUVWXZKRYPTO
C4: CDEFGHIJLMNQUVWXZKRYPTOSAB
C5: IJLMNQUVWXZKRYPTOSABCDEFGHI
C6: SABCDEFHGHIJLMNQUVWXZKRYPTO
C7: SABCDEFHGHIJLMNQUVWXZKRYPTO
C8: ABCDEFGHIJLMNQUVWXZKRYPTOS

VFPJUDEE HZWETZYV GWHKKQET GFQJNCEG GWHKK?DQM
ITWASTOT ALLYINVI SIBLEHOW STHATPOS SIBLE?THE

CPFQZDQM MIAGPFXH QRLGTIMV MZJANQLV KQEDAGDV
YUSEDTHE EARTHSMA GNETICFI ELDXTHEI NFORMATI

FRPJUNGE UNAQZGZL ECGYUXUE ENJTBJLB QCETBJDF
ONWASGAT HEREDAND TRANSMIT TEDUNDER GROUNDTO

HRRYZET KZEMVDUF KSJHKFWH KUWQLSZF TIHHDDDU
ANUNKNOW NLOCATIO NXDOESLA NGLEYKNO WABOUTTH

VH?DWKBFU FPWNTDKI YCUQZERE EVLDKFEZ MOQQJLTT
IS?THEYSH OULDITSB URIEDOUT THERESOM EWHEREXW

UGSYQPFE UNLAVIDX FLGGTEZ?F KZBSFDQV GOGIPUF
HOKNOWST HEEXACTL OCATION?O NLYWWTHI SWASHISL

HHDRKFFH QNTGPUAE CNUVPDJM QCLQUMUN EDFQELZZ
ASTMESSA GEXTHIRT YEIGHTDE GREESFIF TYSEVENM

VRRGKFFV OEEXBDMV PNFXQEZL GREDNQFM PNZGLFLP
 INUTESSI XPOINTFI VESECOND SNORTHSE VENTYSEV

MRJQYALM GNUVPDXV KPDQUMZB EDMHDAFM JGZNUPLG
 ENDEGREE SEIGHTMI NUTESFOR TYFOURSE CONDSWES

EWJLLAET G
 TIDBYROW S

~~U-187~~ ~~(FOUO)~~ Here is a more readable version, with punctuation added:

"IT WAS TOTALLY INVISIBLE. HOW'S THAT POSSIBLE? THEY USED THE EARTH'S MAGNETIC FIELD. THE INFORMATION WAS GATHERED AND TRANSMITTED UNDERGROUND TO AN UNKNOWN LOCATION. DOES LANGLEY KNOW ABOUT THIS? THEY SHOULD. ITS BURIED OUT THERE SOMEWHERE. WHO KNOWS THE EXACT LOCATION? ONLY W.W. THIS WAS HIS LAST TRANSMISSION. THIRTY-EIGHT DEGREES, FIFTY-SEVEN MINUTES, SIX POINT FIVE SECONDS NORTH. SEVENTY-SEVEN MINUTES, FORTY-FOUR SECONDS WEST. I.D. BY ROWS."

~~U-187~~ ~~(FOUO)~~ The reference to W.W. is presumed to be William Webster, former director of the CIA. The coordinates given are a location within the CIA grounds, most likely the main complex or the courtyard area. The meaning of "I.D. BY ROWS" is not known at this time. The repeating key of **ABSCISSA** is defined by Webster's New World Dictionary as, "the horizontal Cartesian coordinate on a plane, measured from the y-axis along a line parallel with the x-axis to point P".

~~U-187~~ After reading this section, it became apparent that the sculpture contained a minimum of 4 parts with one preceding this polyalphabetic section (the first two lines), and two following. More on that a little later.

~~CONFIDENTIAL//SI~~(b)(1)
(b)(3)-50 USC 403
(b)(3)-P.L. 86-36
(b)(3)-18 USC 798**THE SECOND BREAKTHROUGH**

~~(C//SI)~~ ~~(S)~~ As stated earlier, the section that began at line 15 and most likely finished at line 25 (now referred to as section 3) was a probable transposition system. [REDACTED]

[REDACTED] Nonetheless, analysis continued.

~~(S)~~ The attack that was eventually successful on this portion was one of cribbing by brute force. It was noted that there was a single occurrence of the letter **Q**, and just 5 occurrences of the letter **U**, a likely combination. The 3 or 4 letters surrounding the **Q** were paired with the corresponding letters surrounding each **U**. The results were as follows:

Y N	Y T	Y F	Y T	Y F
T R	T E	T I	T U	T M
E E	E N	E O	E E	E F
Y Y	Y E	Y T	Y T	Y E
Q U	Q U	Q U	Q U	Q U
H L	H D	H E	H A	H H
E D	E R	E T	E E	E E
E S	E E	E U	E O	E E
N L	N T	N A	N T	N C

~~(S)~~ Though many wrong turns were taken, the third pair of columns proved correct and the letter **T** was placed in front of the **HE**, produced the word **THE**. The best combination that matched with the remainder was this:

L Y F
E T I
H E O
A Y T
G **Q U**
T H E
D E T
H E U
A N A

~~(S)~~ Many of the trigraphs above looked plausible. Using the letter **G** in front of the **QU**, pairs of columns were cribbed to form **INGQU**, with the following alignment yielding the best looking results:

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~~SECRET~~

O W L Y F
A M E T I
G T H E O
R W A Y T
I N G Q U
E D T H E
W I D E T
F T H E U
E E A N A

⁵⁸
(S) Continuing to use column matching with the remainder of the cipher, a solution was discovered that used an incompletely filled 4 X 86 matrix. That solution follows, but because the matrix is 86 columns wide, the representation has been split into two pieces:

SLOWLYDESPARATLYSLOWLYTHEREMAINSOFPASSAGEDE
ASREMOVEDWITHTREMBLINGHANDSIMADEATINYBREACH
OLEALITTLEIINSERTEDTHECANDLEANDPEEREDINTHEH
FLICKERBUTPRESENTLYDETAILSOFTHEROOMWITHINEM

BRISTHATENCUMBEREDTHELOWERPARTOFTHEDOORWAYW
INTHEUPPERLEFTHANDCORNERANDTHENWIDENINGTHEH
OTAIRESCAPINGFROMTHECHAMBERCAUSEDTHEFLAMETO
ERGEDFROMTHEMISTXCANYOUSEEANYTHINGQ

¹¹ ~~18~~ Further, a very logical key was discovered. The cipher that started this section was from line 15:

E N D Y A H R O H N L S R H E O C P T E O I B I D Y S H N A I

This section can be located in the recovered plaintext, and is inscribed in the matrix from bottom to top, beginning near the end of the message and proceeding towards the front of the message.

SLOWLYDESPARATLYSLOWLYTHEREMAINSOFPASSAGEDE
ASREMOVEDWITHTREMBLINGHANDSIMADEATINYBREACH
OLEALITTLEIINSERTEDTHECANDLEANDPEEREDINTHEH
FLICKERBUTPRESENTLYDETAILSOFTHEROOMWITHINEM

1	1	1	9	8	7
2	1	0			

BRISTHATENCUMBEREDTHELOWERPARTOFTHEDOORWAYW
INTHEUPPERLEFTHANDCORNERANDTHENWIDENINGTHEH
OTAIRESCAPINGFROMTHECHAMBERCAUSEDTHEFLAMETO
ERGEDFROMTHEMISTXCANYOUSEEANYTHINGQ

6 5 4 3 2 1

~~(S)~~ Proceeding backwards in this manner and labeling each column with a number, it becomes obvious that there is indeed a "method to the madness". What follows is the label for each column written out in order, on a width of 7:

```

49 12 61 24 73 36 85
48 11 60 23 72 35 84
47 10 59 22 71 34 83
.. .. .. .. ..
.. .. .. .. ..
39 02 51 14 63 26 75
38 01 50 13 62 25 74
37 86

```

~~(S)~~ Looked at this way, the patterns within each of the 7 columns are plain, with the possible exception of the "86" at the bottom of the second column. Further, a common procedure in columnar transposition systems is to extract columns of the matrix in an order determined by a specific key, often denoted by a keyword to make the key easy to remember. To generate a numerical key based on a keyword of **KRYPTOS** for example, number the keyword based on alphabetical order:

```

K R Y P T O S
1 4 7 3 6 2 5

```

~~(S)~~ Note how a variation of **KRYPTOS** (spelled backwards and wrapped around the ends) fits the columns of the matrix:

```

R K S O T P Y
4 1 5 2 6 3 7
49 12 61 24 73 36 85
48 11 60 23 72 35 84
47 10 59 22 71 34 83
.. .. .. .. ..
.. .. .. .. ..
39 02 51 14 63 26 75
38 01 50 13 62 25 74
37 86

```

~~U (FOUO)~~ Although this may or may not be the way the system was originally set up by Sanborn, it is likely to be very close to the truth. The entire text of section 3 follows, with appropriate punctuation:

"SLOWLY DESPARATLY SLOWLY THE REMAINS OF PASSAGE DEBRIS THAT ENCUMBERED THE LOWER PART OF THE DOORWAY WAS REMOVED. WITH TREMBLING HANDS I MADE A TINY BREACH IN THE UPPER LEFT HAND CORNER, AND THEN, WIDENING THE HOLE A LITTLE, I INSERTED THE CANDLE AND PEERED IN. THE HOT AIR ESCAPING FROM THE CHAMBER CAUSED THE FLAME TO FLICKER, BUT PRESENTLY, DETAILS OF THE ROOM WITHIN EMERGED FROM THE MIST X CAN YOU SEE ANYTHING Q"

~~U (FOUO)~~ In the first line, **DESPERATELY** was misspelled. In the last line, the **X** acts as a period, though other areas of the text should have used periods as well. The **Q** at the end of the text appears to act as a question mark, though the sculpture had a question mark that delineated the end of this section. Earlier, it was noted that in line 15 the **Y**, **A**, and **R** were slightly raised in relation to other surrounding letters. That had no affect on reading this section.

~~U (FOUO)~~ If you have ever read about King Tut, the passage may have sounded familiar to you. It is a paraphrasing from the book "**The Tomb of Tut-ankh-amen**" written by Howard Carter.

THE THIRD BREAKTHROUGH

~~56~~ (S) The last portion read was the first 2 lines of the cipher, section 1. A statistical analysis revealed that this section had width properties that were significant, similar to the first breakthrough, but this time on a width of 5, again implying a polyalphabetic system. Assuming another polyalphabetic system using 5 alphabets, the frequency count follows:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	-	-	2	1	-	-	2	-	-	-	1	2	-	-	-	2	-	-	-	2	-	-	-	-	-
-	-	-	-	-	1	1	-	-	3	2	1	1	-	-	-	-	-	-	1	1	-	1	-	1	-
-	-	-	1	-	-	-	-	-	1	1	-	1	-	-	-	1	-	1	1	-	-	1	5	-	-
-	-	-	-	-	2	-	-	-	-	-	-	1	-	-	3	3	-	-	2	-	-	-	-	1	-
-	1	-	1	1	1	-	-	1	-	-	1	-	-	1	-	-	2	2	-	1	-	-	-	-	-

~~56~~ (S) As before, it was hoped that you could place another alphabet on top of these frequency counts, hopefully resulting in something that would match up nicely. Since the keyword-mixed alphabet based on **KRYPTOS** was used before, we expected that it might be a good choice here. Therefore, a frequency count based on that assumption is given below:

K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
-	-	-	-	-	-	-	1	-	-	2	1	-	-	2	-	-	1	2	-	2	-	2	-	-	-
2	-	-	-	-	-	-	-	-	-	-	1	1	-	-	3	1	1	-	-	1	1	-	1	1	-
1	1	5	-	1	-	-	-	-	-	1	-	-	-	-	-	1	-	1	-	1	-	-	1	-	-
-	3	-	-	-	-	-	-	-	-	-	2	-	-	-	-	-	1	3	2	-	-	-	-	1	-
-	-	-	1	2	-	2	-	1	-	1	1	1	-	-	1	-	1	-	-	-	-	1	-	-	-

~~56~~ (S) The small number of characters made solution difficult in either case, but as before, analysis continued. Beginning with a likely starting point, the third alphabet with cipher value **Y** appearing 5 times was assumed to be a plain value **E**. Due to a higher occurrence of the more common letters, the output appeared to be better using the **KRYPTOS** alphabet, which results in the following:

EMUFP HZLRF AXYUS DJKZL DKRNS HGNFI VJYQT QUXQB
 ..T.. ..S.. ..E.. ..I.. ..D.. ..B.. ..E.. ..G..

QVYUV LLTRE VJYQT MKYRD MFD
 ..E.. ..N.. ..E.. ..L.. ..N

~~56~~ (S) Through trial and error, another **KRYPTOS** alphabet was placed against the frequencies in the first alphabet, which

yielded the following reasonable text patterns:

EMUFP HZLRF AXYUS DJKZL DKRNS HGNFI VJYQT QUXQB
B.T.. E.K.. T.E.. A.C.. A.D.. E.Y.. N.E.. L.A..

QVYUV LLTRE VJYQT MKYRD MFD
L.E.. H.G.. N.E.. I.E.. I.N

~~U (S)~~ With those recoveries in place, further progress was soon made. It was noted that if the last 3 letters of plain were **ION**, then cipher VJYQT (which occurs twice) would become plain **NCE..**

EMUFP HZLRF AXYUS DJKZL DKRNS HGNFI VJYQT QUXQB
BET.. EMK.. TLE.. ACC.. AND.. ESY.. NCE.. LHA..

QVYUV LLTRE VJYQT MKYRD MFD
LIE.. HDG.. NCE.. INE.. ION

~~U (S)~~ While portions of the text seemed to have good recoveries, such as **AND**, **NCE**, **LIE**, and **ION**, other portions appeared less encouraging, like **EMK**, and **HDG**. Further scrutiny revealed that the "good" recoveries occurred in every other group of 5 letters. Using 10 alphabets instead of 5 would eliminate the "bad" recoveries. Looking back at the width statistics, width 10 had also scored high, and the width of 5 was probably a reflection of that.

EMUFPHZLRF AXYUSDJKZL DKRNSHGNFI VJYQTQUXQB
BET..... TLE..... AND..... NCE.....

QVYUVLLTRE VJYQTMKYRD MFD
LIE..... NCE..... ION

~~U (S)~~ Evaluating this new approach, it seemed that the end of the text should be **TION** or **SION**, and that the 2 occurrences of **NCE** would be preceded by a vowel. The following alignment of the tenth alphabet provided that:

EMUFPHZLRF AXYUSDJKZL DKRNSHGNFI VJYQTQUXQB
BET.....B TLE.....G AND.....E NCE.....T

QVYUVLLTRE VJYQTMKYRD MFD
LIE.....A NCE.....S ION

~~(S)~~ Notice that the 2 occurrences of **NCE** are preceded by **E** and **A**, exactly as expected. Further cribbing ensued, which eventually yielded the following set of alphabets and the following decryption:

P: K R Y P T O S A B C D E F G H I J L M N Q U V W X Z
C1: P T O S A B C D E F G H I J L M N Q U V W X Z K R Y
C2: A B C D E F G H I J L M N Q U V W X Z K R Y P T O S
C3: L M N Q U V W X Z K R Y P T O S A B C D E F G H I J
C4: I J L M N Q U V W X Z K R Y P T O S A B C D E F G H
C5: M N Q U V W X Z K R Y P T O S A B C D E F G H I J L
C6: P T O S A B C D E F G H I J L M N Q U V W X Z K R Y
C7: S A B C D E F G H I J L M N Q U V W X Z K R Y P T O
C8: E F G H I J L M N Q U V W X Z K R Y P T O S A B C D
C9: S A B C D E F G H I J L M N Q U V W X Z K R Y P T O
C10: T O S A B C D E F G H I J L M N Q U V W X Z K R Y P

EMUEP HZLRF AXYUS DJKZL DKRNS HGNFI VJYQT QUXQB
BETWE ENSUB TLESH ADING ANDTH ABSCE NCEOF LIGHT

QVYUV LLTRE VJYQT MKYRD MFD
LIEST HENUA NCEOF IQLUS ION

(FOUO) Respaced and punctuated, it reads:

"BETWEEN SUBTLE SHADING AND THE ABSENCE OF LIGHT LIES THE NUANCE OF ILLUSION"

~~(FOUO)~~ In the original decrypt, the word **ILLUSION** was misspelled as **IQLUSION**. The source of this quote is currently unknown. The repeating key of **PALIMPSEST**, below the index letter **K**, has a very interesting definition when viewed in conjunction with the sculpture. It is defined by Webster's New World Dictionary as, "a parchment, tablet, etc., that has been written upon or inscribed two or three times, the previous text or texts having been imperfectly erased and remaining, therefore, still partly visible". Another definition from Webster's Third New International Dictionary is "a memorial brass having earlier engraving on the side opposite to that which is exposed".

THE FOURTH BREAKTHROUGH??

~~S6~~ (S) Unfortunately, a fourth breakthrough has not yet occurred. There are only 97 characters remaining in section 4, but the first section contained just 63 characters and was exploited, meaning a solution is certainly possible, depending on the cryptosystem. A statistical analysis of this portion showed some roughness on interval 7. This could be a characteristic of plaintext auto-key, if the alphabet used has a high frequency letter assigned the value of 0. Another hypothesis is that this last section employs both of the systems already used. First the message is encrypted using some set of alphabets, as was done in the first and third breakthroughs, and then the cipher is put through a transposition, such as that used in the second breakthrough. If the original text had a repeat at a distance of 7 apart (or perhaps 14 or even 21 apart), then after transposing the text, the repeat would now show up in the interval statistic rather than the width statistic.

~~U~~ ~~(S)~~ There is no solution at the current time, although some attempts have been made using plaintext auto-key and other attempts using transposed substitution as the enciphering mechanism.

RECAP

~~U (FOUO)~~ The first section that reads are the first 2 full lines of cipher, a total of 63 characters. The cryptography is a periodic polyalphabetic substitution system employing 10 alphabets. The plain and cipher components are both a keyword mixed sequence based on **KRYPTOS**, using a repeating key of **PALIMPSEST** below the index letter **K**. The plaintext reads:

"BETWEEN SUBTLE SHADING AND THE ABSENCE OF LIGHT LIES THE NUANCE OF ILLUSION"

~~U (FOUO)~~ The second part reads using the cipher from lines 3-14, a total of 370 characters. The cryptography is another periodic polyalphabetic substitution, employing 8 alphabets. The plain and cipher components are both a keyword mixed sequence based on **KRYPTOS**, using a repeating key of **ABSCISSA** below the index letter **K**. The plaintext reads:

"IT WAS TOTALLY INVISIBLE. HOW'S THAT POSSIBLE? THEY USED THE EARTH'S MAGNETIC FIELD. THE INFORMATION WAS GATHERED AND TRANSMITTED UNDERGROUND TO AN UNKNOWN LOCATION. DOES LANGLEY KNOW ABOUT THIS? THEY SHOULD. ITS BURIED OUT THERE SOMEWHERE. WHO KNOWS THE EXACT LOCATION? ONLY W.W. THIS WAS HIS LAST TRANSMISSION. THIRTY-EIGHT DEGREES, FIFTY-SEVEN MINUTES, SIX POINT FIVE SECONDS NORTH. SEVENTY-SEVEN MINUTES, FORTY-FOUR SECONDS WEST. I.D. BY ROWS."

~~U (FOUO)~~ The third section uses the cipher contained in lines 14 through the question mark in line 25. The cryptography is a keyed columnar transposition. The matrix is an incompletely filled 4 X 86, using a key of **KRYPTOS** that has been numerically keyed and repeated 13 times. The plaintext reads:

"SLOWLY DESPARATELY SLOWLY THE REMAINS OF PASSAGE DEBRIS THAT ENCUMBERED THE LOWER PART OF THE DOORWAY WAS REMOVED. WITH TREMBLING HANDS I MADE A TINY BREACH IN THE UPPER LEFT HAND CORNER, AND THEN, WIDENING THE HOLE A LITTLE, I INSERTED THE CANDLE AND PEERED IN. THE HOT AIR ESCAPING FROM THE CHAMBER CAUSED THE FLAME TO FLICKER, BUT PRESENTLY, DETAILS OF THE ROOM WITHIN EMERGED FROM THE MIST. CAN YOU SEE ANYTHING?"

~~U (FOUO)~~ The fourth part has not been read, but most likely uses the last 4 characters of line 25, as well as the cipher in lines 26-28, for a total of 97 characters. It is highly probable

that **KRYPTOS** plays an integral part in the solution, as it did in the 3 parts that have been exploited.